

Tasmia Hossain

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SKILLS

Languages: C, C++, Java, Python

Web & Frameworks: HTML, CSS, Bootstrap, ReactJS, JavaScript, .NET Core

Databases: MySQL, Microsoft SQL Server, MongoDB

Mobile Development: Flutter, Dart

AI/ML & Research: Graph Neural Networks, TinyML, NLP, Jupyter Notebook, Google Colab

Tools: Git, GitHub, VS Code, Android Studio, XAMPP

RESEARCH

Traffic Flow Prediction and Dynamic Routing in Bangladesh's Urban Transport Networks with Graph Neural Networks

Ongoing

- Developing a GNN-based deep learning model to predict urban traffic flow patterns and congestion hotspots across Dhaka's road networks.
- Designing real-time adaptive routing algorithms on graph-structured transport data for smart city congestion management.

PROJECTS

Lost and Found Hub

2025

ASP.NET Core MVC, C#, SQL Server, SignalR, Bootstrap [\[GitHub\]](#)

- Built a full-stack web platform for lost-item recovery with automated image-based matching, real-time SignalR notifications, and ASP.NET Identity authentication.
- Implemented SMTP email alerts and advanced search filters by category, date, and location across a complete MVC architecture.

SafeParking

2024

PHP, MySQL, JavaScript, Bootstrap, Google Maps API [\[GitHub\]](#)

- Developed a web application connecting car owners with landlords for urban parking; features subscription contracts, booking management, and admin dashboard.
- Integrated Google Maps API for geolocation-based spot discovery; implemented ratings, reviews, and payment history tracking.

Inventory Control System

2024

PHP, MySQL, JavaScript, Bootstrap [\[GitHub\]](#)

- Built a food-industry inventory management system with role-based access control (admin/client/supplier), product & brand management, and supplier tracking.
- Developed analytics and reporting module for order history across configurable date ranges.

Sudoku Bliss

2023

Flutter, Dart, SQLite [\[GitHub\]](#)

- Developed a cross-platform mobile Sudoku game with algorithmic puzzle generation, variable difficulty modes, save/resume functionality, and SQLite persistence.
- Delivered polished UX with real-time timer, error/move counters, and in-app tutorials for new players.

EDUCATION

Ahsanullah University of Science and Technology

B.Sc. in Computer Science & Engineering

Dhaka, Bangladesh

CGPA: 3.63 / 4.00 | Expected June 2026

Dhaka City College

Higher Secondary Certificate (HSC)

Dhaka, Bangladesh

GPA: 5.00 / 5.00 | 2020

YWCA Higher Secondary Girls' School

Secondary School Certificate (SSC)

Dhaka, Bangladesh

GPA: 5.00 / 5.00 | 2018

LANGUAGES

Bengali — Native English — Fluent